UNIVERSITY OF PETROLEUM AND ENERGY STUDIES
DEHRADUN

Modeling and Stimulation

Lab Experiment 10

MTECH-COMPUTER SCIENCE ENGINEERING
CYBER SECURITY AND FORENSICS

Name: Jigesh Sheoran
SAP ID: 590025428

Aim

To design and develop a web-based interface for interacting with and visualizing simulation results.

Objectives:

- To understand integration of simulation models with web interfaces
- To create a user-friendly interface using HTML, CSS, and JavaScript
- To allow users to input parameters dynamically
- To visualize simulation results interactively

Theory :

Web-based simulation interfaces allow users to interact with computational models through graphical user interfaces. This enhances usability and accessibility of simulation tools.

In this experiment, a simple simulation (random walk) is implemented and controlled through a web interface. The user provides input parameters such as number of steps, and the simulation results are visualized dynamically.

Technologies used:

- **HTML** → Structure
- **CSS** → Styling
- **JavaScript** → Simulation logic + interaction

Methodology:

2.1 Interface Design

- Input fields for simulation parameters
- Button to trigger simulation
- Canvas/graph area for visualization

2.2 Simulation Integration

- JavaScript handles random walk logic
- Input values are passed dynamically

2.3 Visualization

- Results plotted using canvas

Algorithm:

1. Take user input (number of steps)
2. Initialize position = 0
3. For each step:
    - **Move +1 or -1 randomly**
4. Store positions
5. **Plot the result dynamically**

Code:

```
<!DOCTYPE html>
<html>
<head>
    <title>Random Walk Simulation</title>
    <style>
        body {
            font-family: Arial;
            text-align: center;
            background: #f4f4f4;
        }
        canvas {
            border: 1px solid black;
            margin-top: 20px;
        }
        input, button {
            padding: 10px;
            margin: 5px;
        }
    </style>
</head>

<body>

<h2>Random Walk Simulation</h2>

<input type="number" id="steps" placeholder="Enter number of steps">
<button onclick="runSimulation()">Run Simulation</button>

<br>
<canvas id="graph" width="600" height="300"></canvas>

<script>
function runSimulation() {
    let steps = document.getElementById("steps").value;

    let position = 0;
    let data = [position];

    for (let i = 0; i < steps; i++) {
        let step = Math.random() < 0.5 ? -1 : 1;
        position += step;
        data.push(position);
    }

```

```
    drawGraph(data);
}

function drawGraph(data) {
    let canvas = document.getElementById("graph");
    let ctx = canvas.getContext("2d");

    ctx.clearRect(0, 0, canvas.width, canvas.height);

    let max = Math.max(...data);
    let min = Math.min(...data);

    let scaleX = canvas.width / data.length;
    let scaleY = canvas.height / (max - min + 1);

    ctx.beginPath();
    ctx.moveTo(0, canvas.height / 2);

    for (let i = 0; i < data.length; i++) {
        let x = i * scaleX;
        let y = canvas.height - ((data[i] - min) * scaleY);
        ctx.lineTo(x, y);
    }

    ctx.stroke();
}
</script>

</body>
</html>
```

Output :

Results:

- The web interface successfully accepts user input
- Simulation runs dynamically on button click
- Results are visualized in real-time

Observation Table:

Number of Steps	Behavior of Random Walk	Trend Observed
10000	Highly fluctuating, dense path	No clear trend (random)
5000	Downward drift observed	Temporary negative trend
1000	Upward drift observed	Temporary positive trend
500	Moderate fluctuations	Slight upward tendency
250	Noticeable oscillations	No clear trend
100	Limited fluctuations	Balanced around zero
50	Short-term upward movement	Slight positive trend
25	Zig-zag pattern	No clear trend
10	Very small fluctuations	Random behavior
1	Single step movement	Either +1 or -1

Comparison: Expected vs Observed Behavior

Parameter	Expected Behavior (Theory)	Observed Behavior (From Simulation)
Direction of Walk	No fixed direction (equal probability of +1 and -1)	Some runs show upward/downward drift, but no consistent direction
Mean Displacement	Should approach 0 as steps increase	Average remains close to 0 for larger step sizes
Individual Paths	Highly random and unpredictable	Paths show irregular zig-zag and fluctuating patterns
Effect of Increasing Steps	Smoother overall trend due to averaging	Larger step sizes (1000, 5000, 10000) appear smoother and dense
Short-Term Behavior	Can show temporary trends	Small step sizes show visible upward/downward movements
Long-Term Behavior	No long-term drift (unbiased process)	No consistent long-term trend observed
Variability (Variance)	Increases with number of steps	Larger spreads seen in higher step simulations

Conclusion:

The experiment successfully demonstrates the behavior of stochastic processes through random walk simulation using a web-based interface. It was observed that individual random walk paths are highly irregular and may show temporary upward or downward trends; however, no consistent long-term direction exists.

As the number of steps increases, the overall behavior aligns with theoretical expectations, where the average displacement approaches zero. This confirms that the random walk is an unbiased process governed by equal probability of movement in both directions.

The results validate key statistical concepts such as the Law of Large Numbers, where increasing the number of trials reduces randomness in the average outcome, and variance, which explains the spread observed in individual paths.

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